

Image Processing – Fourier Extraction

Finger-Vein PAD using Fourier Magnitude and Phase Features

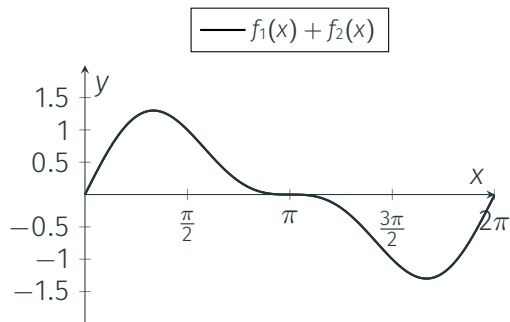
Andreas Auer Philipp Fuchs Sarah Götz

6th February 2026

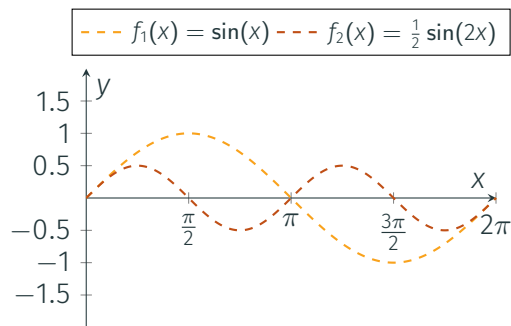
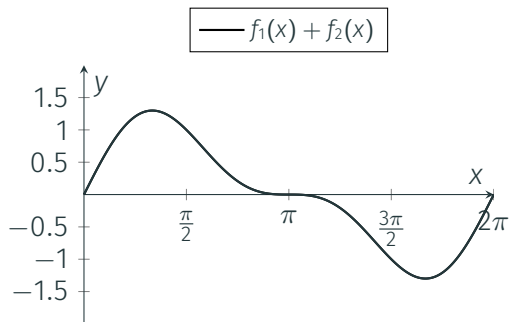
- Fourier Introduction
- Task 1
 - Experiments
 - Results
- Task 2
 - Experiments
 - Results
- Final Results & Conclusion

Fourier Introduction

GRAPHICAL REPRESENTATION OF FOURIER



GRAPHICAL REPRESENTATION OF FOURIER



Fourier-Series

$$f(t) = \sum_{k=-\infty}^{\infty} c_k e^{ik\omega_0 t}$$

- k : index of the frequency components
- $\omega_0 = \frac{2\pi}{T}$: fundamental angular frequency vibration
- $c_k \in \mathbb{C}$: complex fourier coefficients

$$c_k = \operatorname{Re}(c_k) + i \operatorname{Im}(c_k)$$

$$|c_k| = \sqrt{\operatorname{Re}(c_k)^2 + \operatorname{Im}(c_k)^2} \quad (\text{Magnitude})$$

$$\varphi_k = \arg(c_k) \iff c_k = |c_k| e^{i\varphi_k} \quad (\text{Phase})$$

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Magnitude $|c_k|$

→ Strength of the frequency $k \cdot \omega_0$.

Phase φ_k

→ Gives a “position” relative to the x-Axis.

- **Fourier** feature extraction.
 - 2D Fast-Fourier-Transformation (**FFT**)
- FFT returns **N** radial bands → N -dimensional feature vector $\in \mathbb{R}^N$ for:
 1. **Magnitude features:**
 - Power spectrum, summed over each radial band.
 2. **Phase features:**
 - Circular mean of phase values per band.

- L2-normalization:
 - Removes scale differences by projecting onto the unit sphere.
⇒ Improves comparability.
 - Focus on patterns rather than intensity/brightness variations.
 - Always used in our experiments.

- **L2-normalization:**
 - Removes scale differences by projecting onto the **unit sphere**.
⇒ Improves comparability.
 - Focus on **patterns** rather than **intensity/brightness** variations.
 - **Always** used in our experiments.
- **Log-compression (magnitude only):**
 - **Compresses** (“shrinks”) large magnitude spikes
⇒ prevents them from **dominating** the representation.
 - **Spreads** out small magnitude values.
⇒ Subtle details become more visible and **more influential** for learning.
 - Stabilizes features: multiplicative changes → **additive** shifts.
 - **Experiment-based** use.

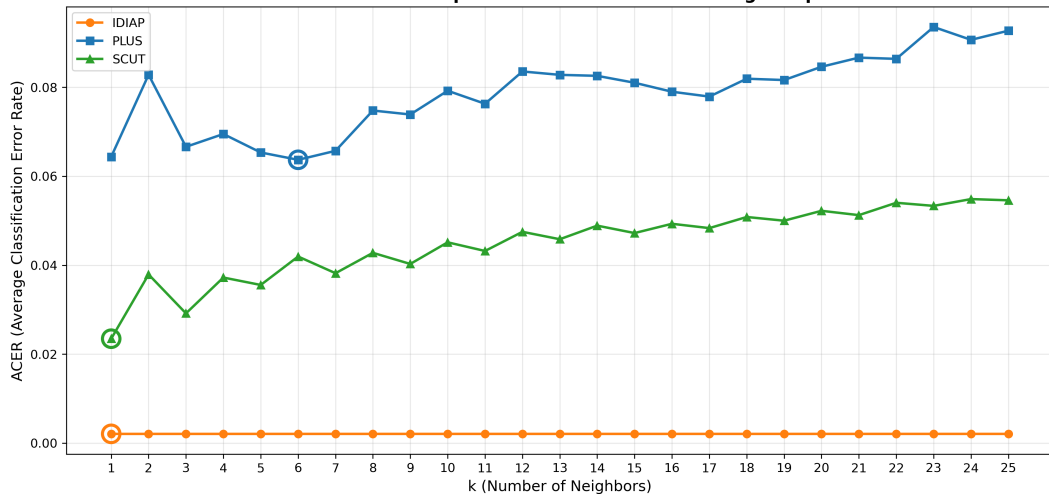
Experiments Task 1

Evaluation criterion: ACER of the baseline configuration

- Optimize k in KNN
 - Search over $k \in [1, 25]$
- Magnitude vs. Phase
 - Evaluate magnitude-only, phase-only, and combined features
- Inter-Class Variation
 - Real vs. Spoof
- Intra-Class Variation
 - Variation within Real and within Spoof samples
- Effect of Log-Compression
 - Compare magnitude features w/ vs. w/o log-compression

EVALUATE BEST k FOR KNN

KNN Performance Comparison: All Datasets WITH Log Compression



- Best k : PLUS: 6, IDIAP: 1, SCUT: 1

Pearson correlation of magnitude and phase feature vectors:

- PLUS: ≈ 0.048
- IDIAP: ≈ -0.010
- SCUT: ≈ 0.016

$\Rightarrow$ Features are almost **uncorrelated** – a **combination** may provide complementary information.

$\Rightarrow$ Let's try it.

$$M = [1, \dots, 10, 11, \dots, 20, 21, \dots, 30]$$

M_1 M_2 M_3

$$P = [1, \dots, 10, 11, \dots, 20, 21, \dots, 30]$$

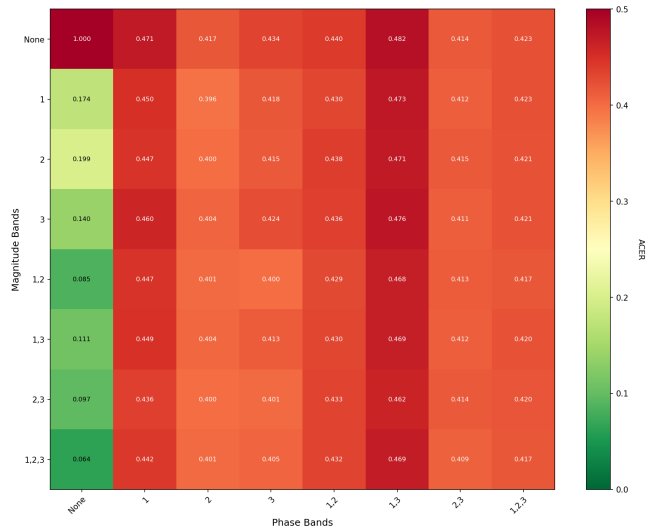
P_1 P_2 P_3

$$\mathcal{P}(M) = \{\emptyset, \{M_1\}, \{M_2\}, \{M_3\}, \{M_1, M_2\}, \{M_1, M_3\}, \{M_2, M_3\}, \{M_1, M_2, M_3\}\}$$

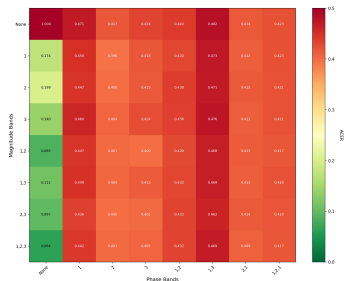
$$\mathcal{P}(P) = \{\emptyset, \{P_1\}, \{P_2\}, \{P_3\}, \{P_1, P_2\}, \{P_1, P_3\}, \{P_2, P_3\}, \{P_1, P_2, P_3\}\}$$

$$\mathcal{P}(M) \times \mathcal{P}(P) = \{\emptyset, \{\emptyset, P_1\}, \{\emptyset, P_2\}, \dots, \{M_1, M_2, M_3, P_1, P_2, P_3\}\}$$

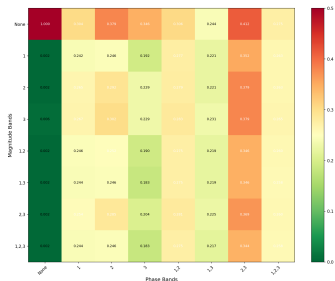
COMBINING DIFFERENT THIRDS OF MAGNITUDE AND PHASE – PLUS DATASET



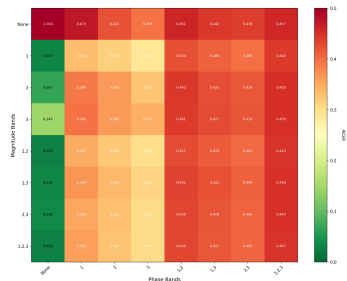
COMBINING DIFFERENT THIRDS OF MAGNITUDE AND PHASE – ALL DATASETS



PLUS



IDIAP

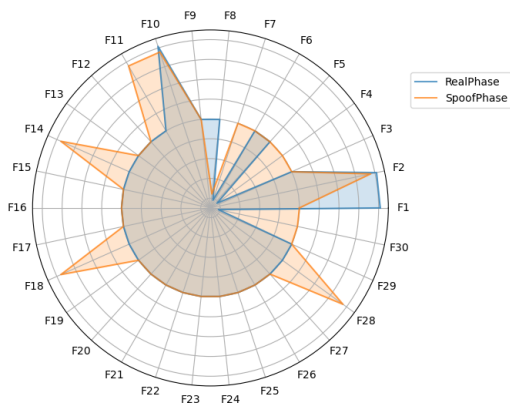
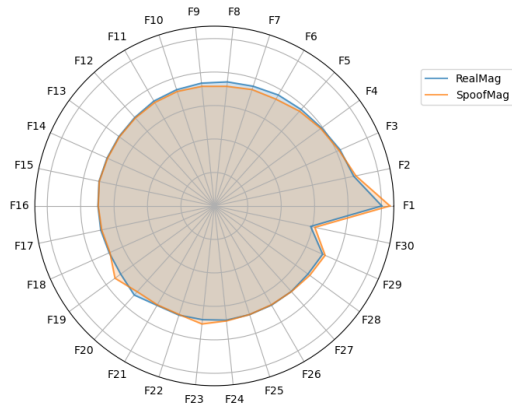


SCUT

⇒ Adding phase severely **degrades** performance.

- Best k : PLUS: 6, IDIAP: 1, SCUT: 1
- **Magnitude only** is best for performance – Phase worsens the results.

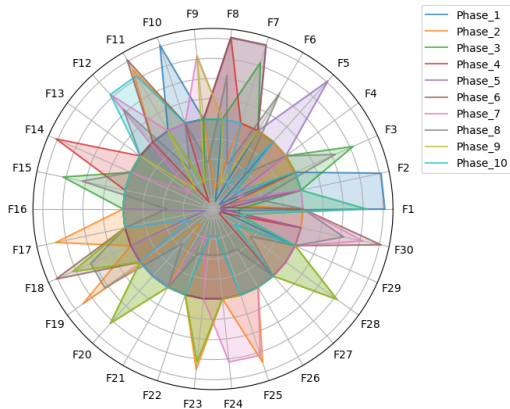
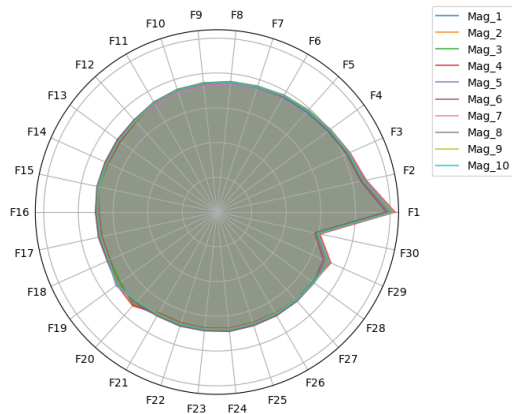
INTER-CLASS VARIATION: REAL VS. SPOOF



⇒ **Magnitude** features: small differences → **low** Inter-Class Variation.

⇒ **Phase** features: large variations → **high** Inter-Class Variation.

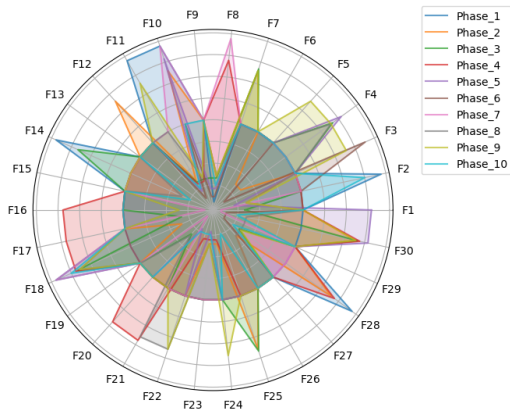
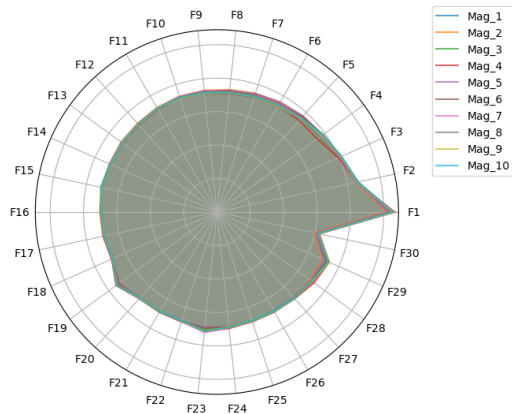
INTRA-CLASS VARIATION: REAL



⇒ **Magnitude** features: very consistent → **low** Intra-Class Variation.

⇒ **Phase** features: highly irregular/random → **high** Intra-Class Variation.

INTRA-CLASS VARIATION: SPOOF

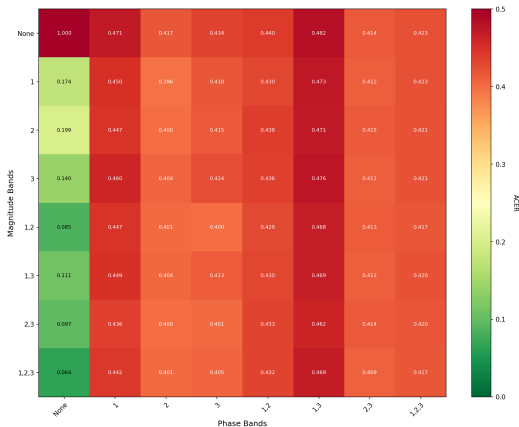


⇒ **Magnitude** features: very consistent → **low** Intra-Class Variation.

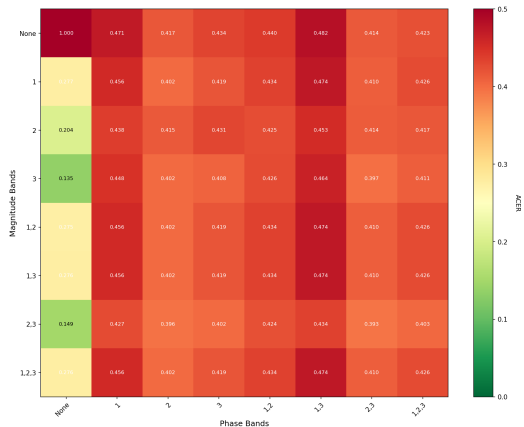
⇒ **Phase** features: highly irregular/random → **high** Intra-Class Variation.

- Best k : PLUS: 6, IDIAP: 1, SCUT: 1
- **Magnitude only** is best for performance – Phase worsens the results.
 - **Magnitude** features:
consistent within each class + small but stable inter-class differences
⇒ good discriminative power.
 - **Phase** features:
large intra-class variation/randomness
⇒ differences between real and spoof not reliably exploitable.

w/ vs. w/o LOG-COMPRESSION – PLUS DATASET



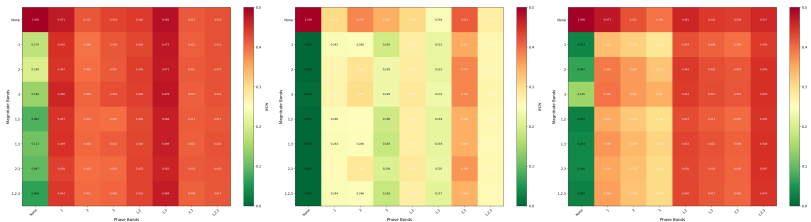
w/ log



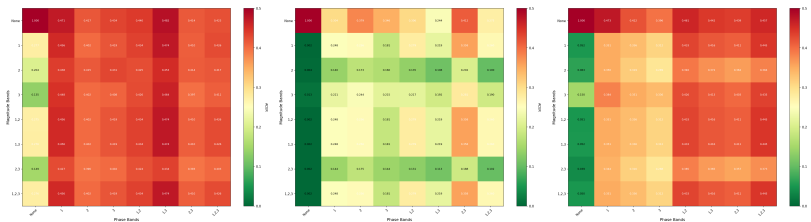
w/o log

w/ vs. w/o LOG-COMPRESSION – ALL DATASETS (PLUS & IDIAP & SCUT)

w/ log



w/o log



⇒ Log-Compression improves results (magnitude-only).

- **Best k :** PLUS: 6, IDIAP: 1, SCUT: 1
- **Magnitude only** is best for performance – Phase worsens the results.
 - **Magnitude** features:
consistent within each class + small but stable inter-class differences
⇒ good discriminative power.
 - **Phase** features:
large intra-class variation/randomness
⇒ differences between real and spoof not reliably exploitable.
- **Log-Compression** improves results in all datasets.

Results Task 1

RESULTS TASK 1 WITH OUR BEST SETTINGS

k: PLUS: 6, IDIAP: 1, SCUT: 1; Magnitude-only; Log-Compression on; Normalization on

Dataset	Baseline (%)			Step 1 (%)			Step 2 (%)		
	APCER	BPCER	ACER	APCER	BPCER	ACER	APCER	BPCER	ACER
IDIAP	0.00	0.42	0.21	0.00	0.42	0.21	0.00	0.00	0.00
PLUS	6.12	6.33	6.23	9.09	6.67	7.88	10.51	13.67	12.09
SCUT	4.31	0.44	2.38	4.86	0.72	2.79	6.69	0.97	3.83

⇒ IDIAP unexpectedly improves,

⇒ PLUS degrades notably,

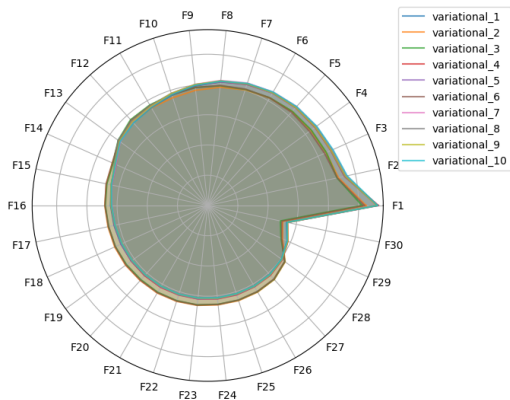
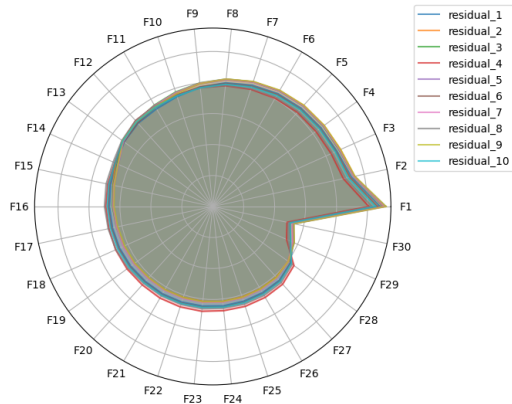
⇒ SCUT degrades slightly.

Experiments Task 2

Evaluation focus: Behaviour of synthetic spoof data

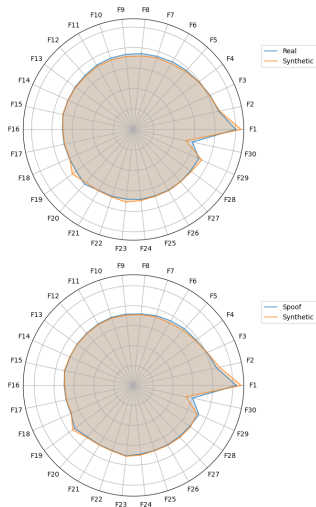
- **Intra-Class** Variation:
 - Synthetic Spoof (Residual)
 - Synthetic Spoof (Variational)
- **Inter-Class** Variation:
 - Real vs. Synthetic Spoof
 - Spoof vs. Synthetic Spoof
- Comparison of Different **Thirds**
 - Are any feature bands better than others?
 - Explaining the results?

INTRA-CLASS VARIATIONS – RESIDUAL & VARIATIONAL



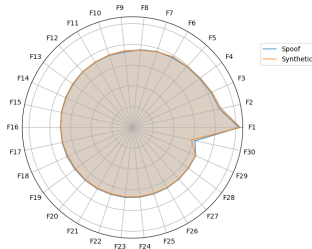
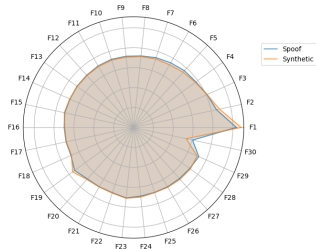
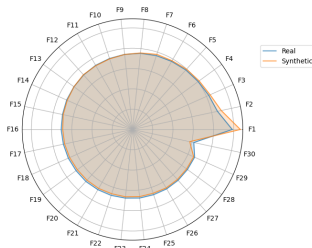
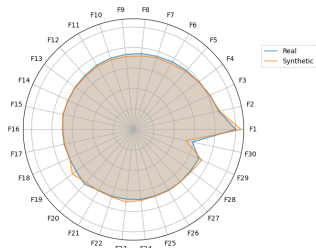
⇒ In both cases: **More** variation than spoof/real images.

INTER-CLASS VARIATIONS – IDIAP & PLUS & SCUT



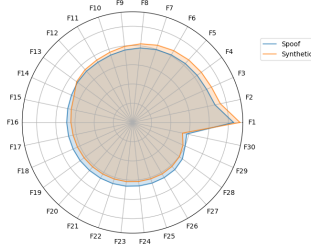
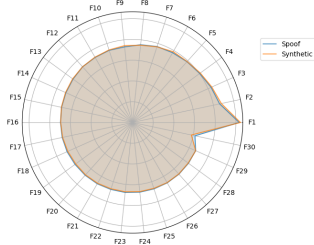
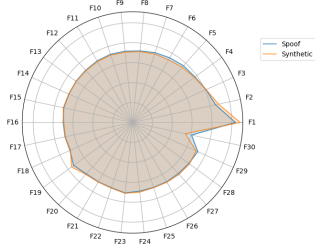
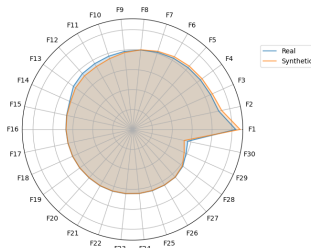
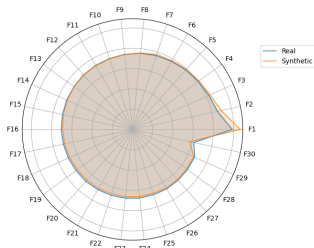
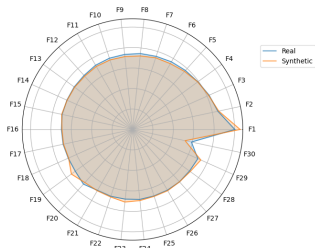
⇒ IDIAP & PLUS: Small difference to Spoofs – recognizable difference to Reals. 25/36

INTER-CLASS VARIATIONS – IDIAP & PLUS & SCUT



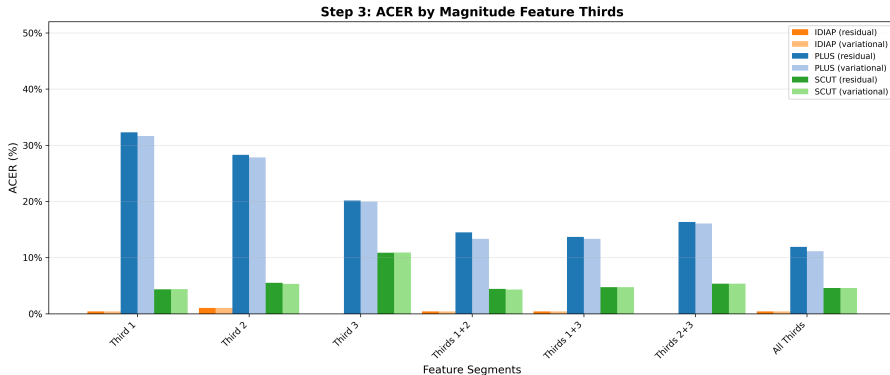
⇒ IDIAP & PLUS: Small difference to Spoofs – recognizable difference to Reals. 25/36

INTER-CLASS VARIATIONS – IDIAP & PLUS & SCUT



⇒ IDIAP & PLUS: Small difference to Spoofs – recognizable difference to Reals. 25/36

COMPARISON OF DIFFERENT THIRDS – STEP 3

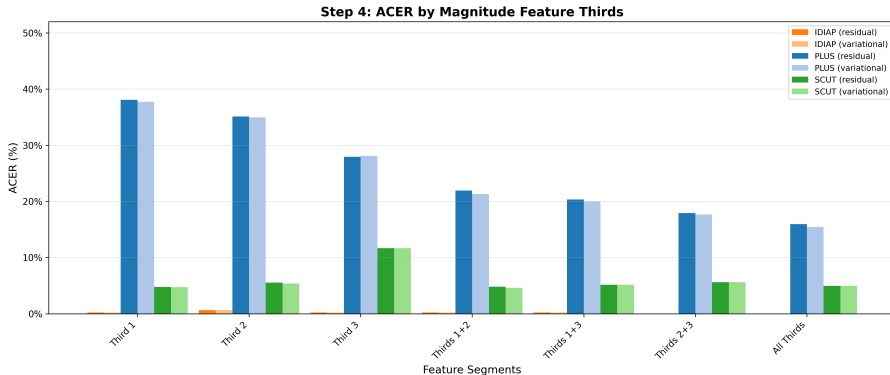


⇒ IDIAP: No notable differences.

⇒ PLUS: “All Thirds” perform best.

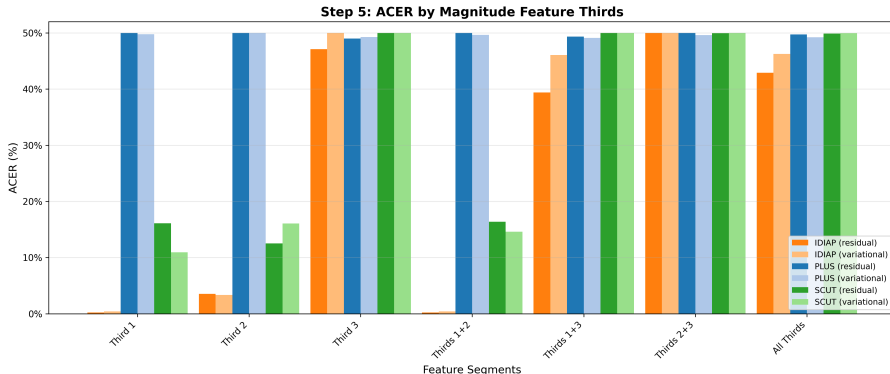
⇒ SCUT: Third 3 only is worse.

COMPARISON OF DIFFERENT THIRDS – STEP 4



- ⇒ IDIAP: No notable differences
- ⇒ PLUS: “All Thirds” perform best
- ⇒ SCUT: Third 3 only is worse

COMPARISON OF DIFFERENT THIRDS – STEP 5

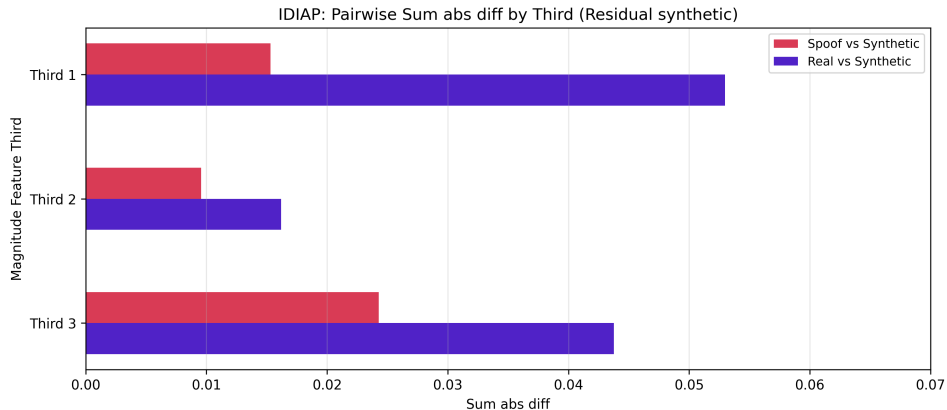


⇒ IDIAP: Third 3 is messing up result → *Overall best: 1+2.*

⇒ PLUS: Bad performance in general → *Overall best: All.*

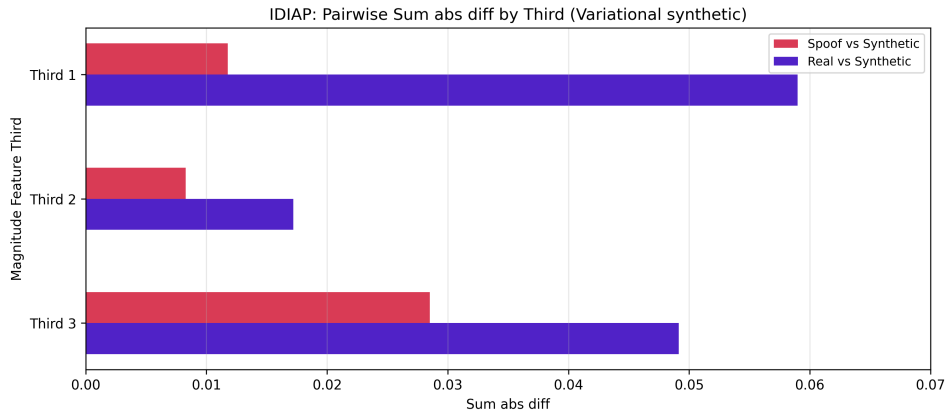
⇒ SCUT: Third 3 is messing up result → *Overall best: 1.*

EXPLANATION – IDIAP – RESIDUAL



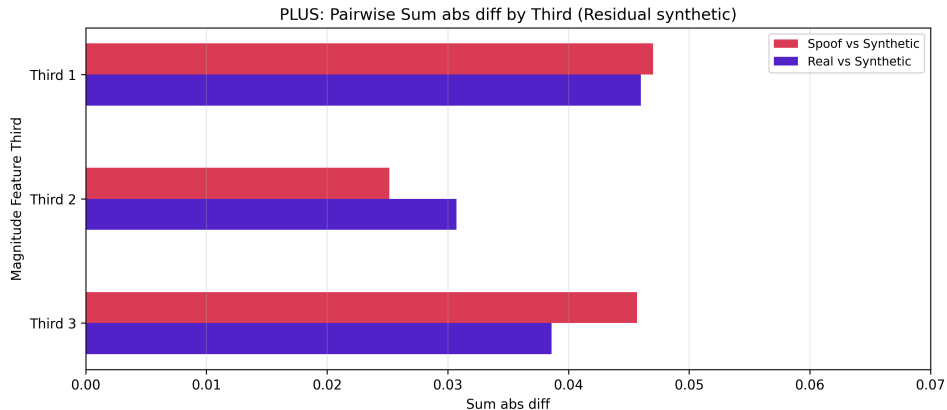
⇒ IDIAP: Third 3 is messing up result → *Overall best: 1+2.*

EXPLANATION – IDIAP – VARIATIONAL



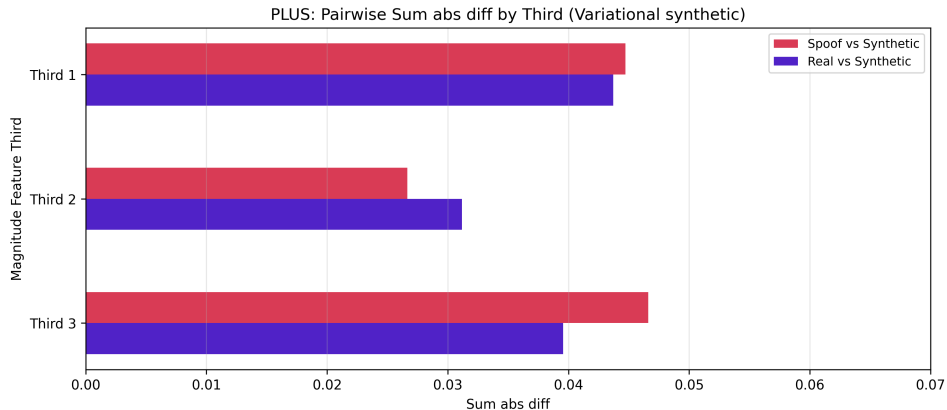
⇒ IDIAP: Third 3 is messing up result → *Overall best: 1+2.*

EXPLANATION – PLUS – RESIDUAL



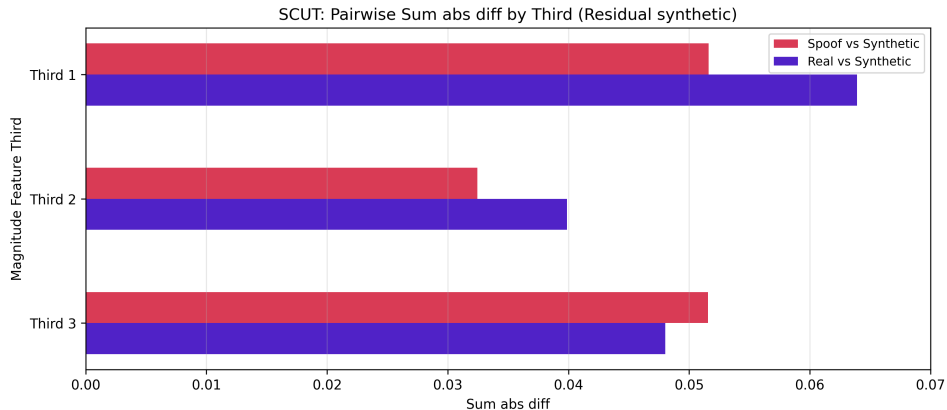
⇒ **PLUS:** Bad performance in general → *Overall best: All.*

EXPLANATION – PLUS – VARIATIONAL



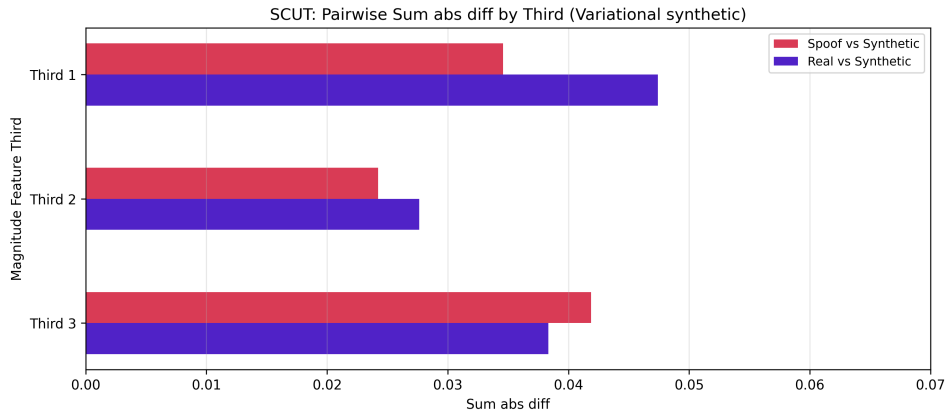
⇒ **PLUS:** Bad performance in general → *Overall best: All.*

EXPLANATION – SCUT - RESIDUAL



⇒ SCUT: Third 3 is messing up result → *Overall best: 1.*

EXPLANATION – SCUT - VARIATIONAL



⇒ **SCUT:** Third 3 is messing up result → *Overall best: 1.*

Results Task 2

RESULTS TASK 2

Top: **Residual** Synthetic Data

Dataset	Thirds	Step 3 (%)			Step 4 (%)			Step 5 (%)		
		APCER	BPCER	ACER	APCER	BPCER	ACER	APCER	BPCER	ACER
IDIAP	12	0.00	0.83	0.42	0.00	0.42	0.21	0.00	0.42	0.21
PLUS	123	19.12	4.67	11.89	29.28	2.63	15.96	99.44	0.00	49.72
SCUT	1	7.86	0.83	4.35	8.97	0.58	4.78	32.14	0.03	16.08

RESULTS TASK 2

Top: **Residual** Synthetic Data – Bottom: **Variational** Synthetic Data

Dataset	Thirds	Step 3 (%)			Step 4 (%)			Step 5 (%)		
		APCER	BPCER	ACER	APCER	BPCER	ACER	APCER	BPCER	ACER
IDIAP	12	0.00	0.83	0.42	0.00	0.42	0.21	0.00	0.42	0.21
PLUS	123	19.12	4.67	11.89	29.28	2.63	15.96	99.44	0.00	49.72
SCUT	1	7.86	0.83	4.35	8.97	0.58	4.78	32.14	0.03	16.08

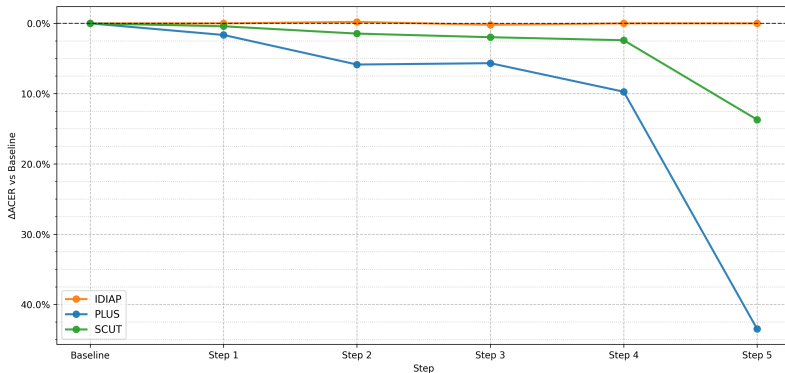
Dataset	Thirds	Step 3 (%)			Step 4 (%)			Step 5 (%)		
		APCER	BPCER	ACER	APCER	BPCER	ACER	APCER	BPCER	ACER
IDIAP	12	0.00	0.83	0.42	0.00	0.42	0.21	0.42	0.42	0.42
PLUS	123	17.57	4.67	11.12	28.28	2.63	15.46	98.44	0.00	49.22
SCUT	1	7.75	1.03	4.39	8.69	0.78	4.74	21.64	0.19	10.92

⇒ **APCER** (false negative) **increases** while BPCER (false positive) decreases.

⇒ Once again, **IDIAP** remains unexpectedly **low** over all steps.

Final Results

FINAL RESULTS



QUESTION: Can synthetic spoof data replace real spoof data?

⇒ Yes, but depending on the datasets only to **some extend**.

Questions?